

AMD / NVDA Statistical Arbitrage

A market-neutral pairs-trading strategy with regime gating, adaptive risk controls, and transparent parameter selection

INDEPENDENT QUANTITATIVE RESEARCH · 2015–2024

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Sample: 2015–2024 · Universe: AMD, NVDA · Frequency: daily

+2.7% CAGR (net)	0.44 Sharpe	−12.9% Max Drawdown	2.81 Profit Factor
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EXECUTIVE SUMMARY

Over a full 2015–2024 sample, a market-neutral AMD/NVDA pairs trade earns a modest but positive net return (+30.1% total, +2.7% CAGR, Sharpe 0.44) while in the market only 8.9% of days. The pair is **not** cointegrated over the full sample; the strategy only trades in the ~15–34% of windows where a trailing cointegration test passes. The headline result is deliberately unembellished: the value here is a disciplined, regime-aware research process — rolling hedge ratio, volatility-adaptive bands, ATR stops, a drawdown circuit-breaker, realistic costs, and a threshold chosen by a pre-committed rule — not an outsized Sharpe.

Full methodology, per-trade log, and the complete eight-iteration research history follow.

1 Introduction & Strategy Logic

Pairs trading is a market-neutral statistical-arbitrage strategy. When two historically related securities drift apart, the strategy takes offsetting long and short positions and profits if the gap (the *spread*) reverts to its historical mean. Because one leg is long and the other short, broad market moves largely cancel, isolating the relative mispricing.

This report tests that idea on AMD and NVDA, two semiconductor names with a strong economic relationship, across a full 2015–2024 daily sample. The central research question is not merely “does it make money” but “*when* is the relationship tradeable, and how much of any result survives realistic risk controls and costs.”

RESEARCH INSIGHT

A full-sample cointegration test on AMD/NVDA **fails** (Engle-Granger $p = 0.94$): the two are not linked over the whole decade. The strategy therefore does not assume a permanent relationship — it re-tests cointegration on a trailing window and only trades when the pair currently passes.

2 Methodology

The engine is fully rules-based. Every parameter is fixed in one configuration block; none was tuned after seeing results, with the single exception of the cointegration threshold, which is selected transparently in Section 4.

Component	Specification
Sample & universe	AMD, NVDA daily OHLC, 2015-01-01 to 2024-12-31 (2,515 trading days)
Cointegration gate	Engle-Granger on a trailing 252-day window, re-checked every 5 days; trade only when $p < 0.20$
Hedge ratio	Rolling 120-day, $\beta = \text{Cov}(\text{AMD}, \text{NVDA}) / \text{Var}(\text{NVDA})$; re-estimated daily (dynamic, not static)
Signal	60-day rolling z-score of the spread ($\text{AMD} - \beta \cdot \text{NVDA}$)
Entry / exit	Volatility-adaptive bands: thresholds scale with spread vol ÷ its trailing 1-year average
Risk controls	ATR-based stop ($3.0 \times 14\text{-day}$), 30-day max hold, 3-day cooldown, 15% drawdown circuit-breaker
Costs	7.0 bps per trade, applied to every entry and exit

Table 1 — Strategy specification. Every tunable value lives in one configuration cell.

METHODOLOGY NOTE

Two design choices matter most. First, the **rolling hedge ratio** lets the dollar-neutral mix adapt as the AMD–NVDA relationship drifts, rather than anchoring to a stale in-sample fit. Second, the **cointegration gate** keeps the strategy flat whenever the pair is not currently mean-reverting, which is most of the time — the strategy holds a position on just 8.9% of days.

3 Cointegration & Regime Analysis

A static, full-sample cointegration test answers only one question for the entire decade, and for AMD/NVDA the answer is negative ($p = 0.94$; full-sample return correlation 0.58). Re-running the test on a rolling 252-day window shows the relationship is intermittent: the pair passes a $p < 0.10$ gate in 14.9% of the sample and a looser $p < 0.30$ gate in 33.7%. Trading is confined to these windows.

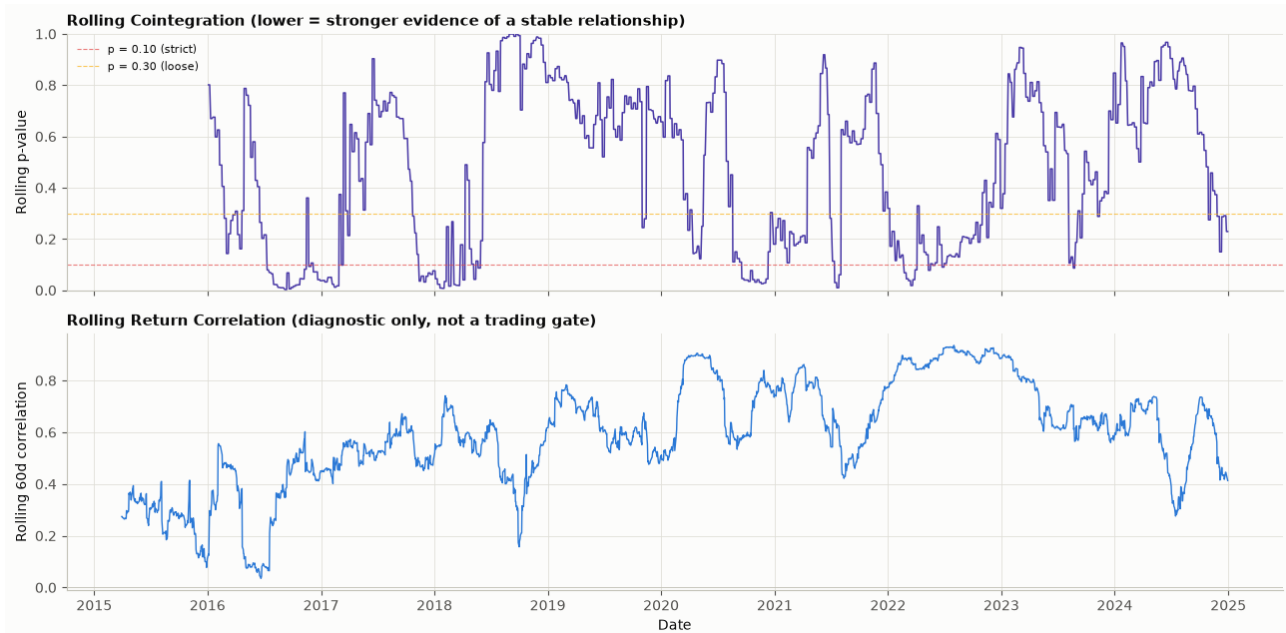


Exhibit A — Rolling cointegration p-value and correlation (notebook output). Shaded/low-p periods are when the strategy is permitted to trade.

WHY THIS MATTERS

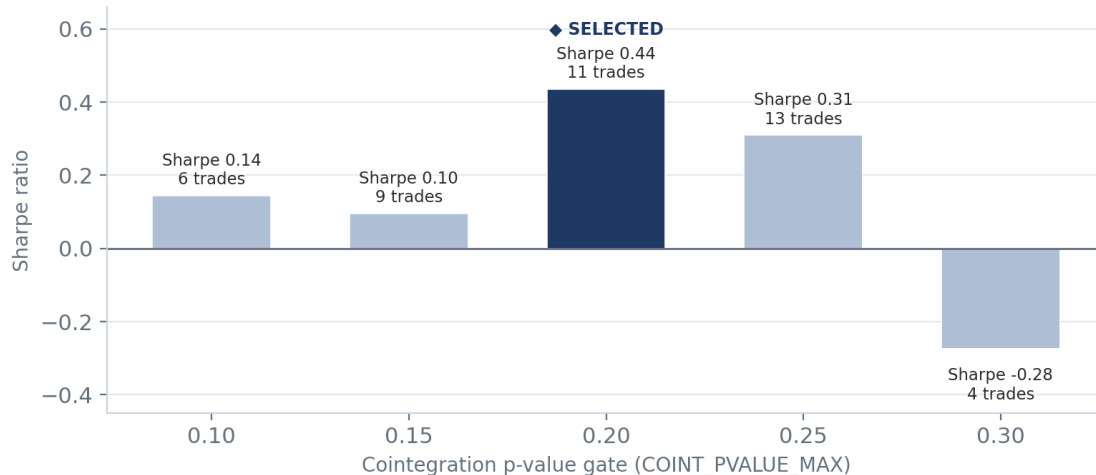
This is the core of the research thesis: AMD/NVDA is not a durable cointegrated pair, so a naive always-on pairs trade is mis-specified. Gating on a live cointegration test converts a losing idea into a modestly positive one by simply refusing to trade when the statistical basis is absent.

4 Sensitivity Analysis: Choosing the Threshold

The cointegration gate (COINT_PVALUE_MAX) is the one genuine judgment call. Rather than tune it silently, the selection rule was fixed *before* viewing results: choose the candidate with the highest Sharpe among those producing at least five completed trades, so the choice is not a two-trade coin flip. That rule selects $p < 0.20$.

Exhibit 3 | Sensitivity Analysis: Threshold Selection

Rule fixed in advance: highest Sharpe among candidates with ≥ 5 trades $\rightarrow p < 0.20$



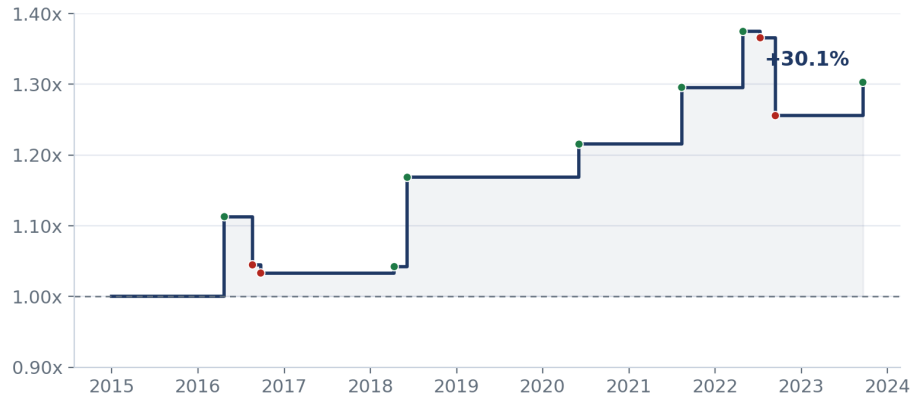
p-gate	Trades	Sharpe	Ann. Return	Max DD	Win Rate
0.10	6	0.15	+0.5%	-8.6%	50.0%
0.15	9	0.10	+0.4%	-12.9%	55.6%
0.20	11	0.44	+2.7%	-12.9%	63.6%
0.25	13	0.31	+1.9%	-14.0%	61.5%
0.30	4	-0.27	-1.3%	-23.1%	25.0%

Table 2 — Sensitivity grid; selected configuration ($p < 0.20$) highlighted. Note $p < 0.25$ trades more but at lower Sharpe, and $p < 0.30$ breaks down entirely.

5 Results

Exhibit 1 | Strategy Equity Curve (by trade, net of costs)

Growth of \$1 compounded across 11 completed trades · reconciles to +30.1% total return



Metric	Value	Metric	Value
Total return (net)	+30.1%	Annualised volatility	6.1%
CAGR (net)	+2.7%	Sharpe ratio	0.44
Max drawdown	-12.9%	Calmar ratio	0.21
Completed trades	11	Trade win rate	63.6%
Profit factor	2.81	Avg trade return	+2.62%
Avg holding period	20.3 days	Days in market	8.9%
Annualised cost drag	0.04%	Daily win rate	50.0%

Table 3 — Full performance metrics, net of 7 bps/trade costs. Sortino is omitted pending a correction: it must be computed over the same daily return series as Sharpe (see Limitations).

Exhibit 2 | Return by Trade

7 of 11 trades profitable (63.6%) · profit factor 2.81 · avg trade +2.62%

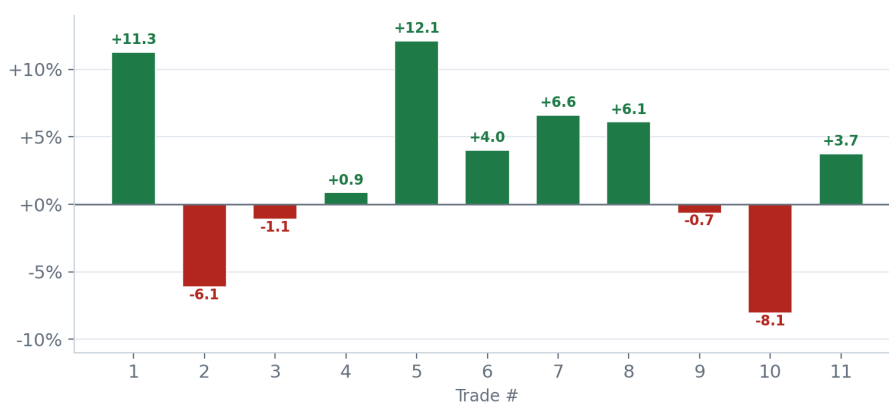


Exhibit 2 — Per-trade net returns; seven of eleven trades profitable.

5.1 Full Backtest Dashboard

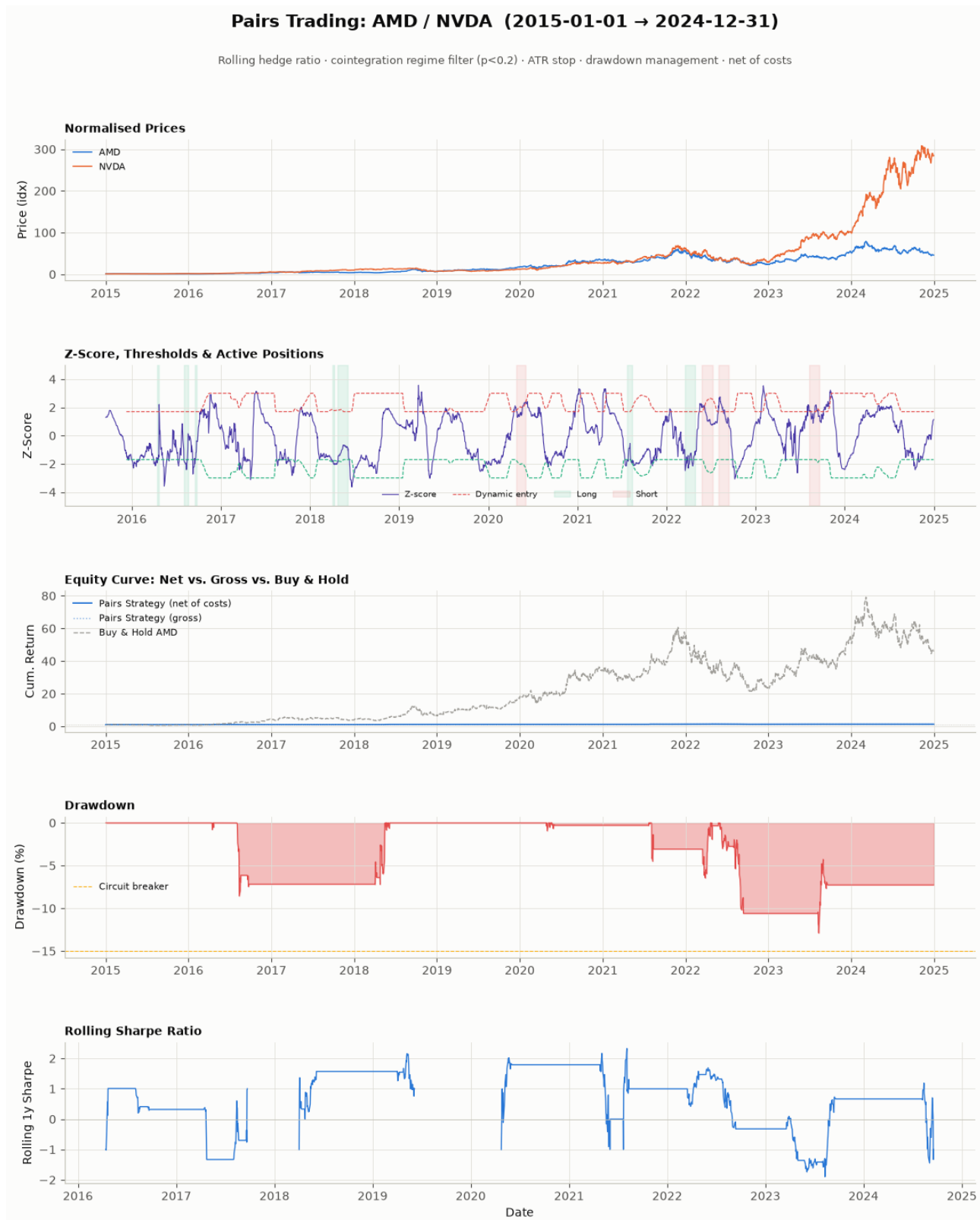


Exhibit B — Prices, z-score with active positions, equity (net/gross vs buy-and-hold), drawdown, and rolling Sharpe (notebook output). The strategy line appears flat against buy-and-hold AMD because both share an axis and AMD compounded ~60× over the

decade; Exhibit 1 shows the strategy on its own scale.

6 Trade Log

#	Entry	Exit	Dir.	Hold (d)	Exit reason	Return
1	2016-04-14	2016-04-22	Long	6	reverted	+11.29%
2	2016-08-02	2016-08-19	Long	13	reverted	-6.12%
3	2016-09-16	2016-09-23	Long	5	reverted	-1.11%
4	2018-04-03	2018-04-12	Long	7	atr-stop	+0.88%
5	2018-04-24	2018-06-06	Long	30	max-hold	+12.12%
6	2020-04-24	2020-06-03	Short	27	atr-stop	+4.01%
7	2021-07-22	2021-08-12	Long	15	atr-stop	+6.61%
8	2022-03-17	2022-04-29	Long	30	max-hold	+6.11%
9	2022-05-25	2022-07-11	Short	30	max-hold	-0.65%
10	2022-08-01	2022-09-13	Short	30	max-hold	-8.06%
11	2023-08-07	2023-09-19	Short	30	max-hold	+3.75%

Table 4 — Complete trade log. Exit reasons: reverted (z-score mean-reversion), atr-stop (volatility stop), max-hold (30-day cap).

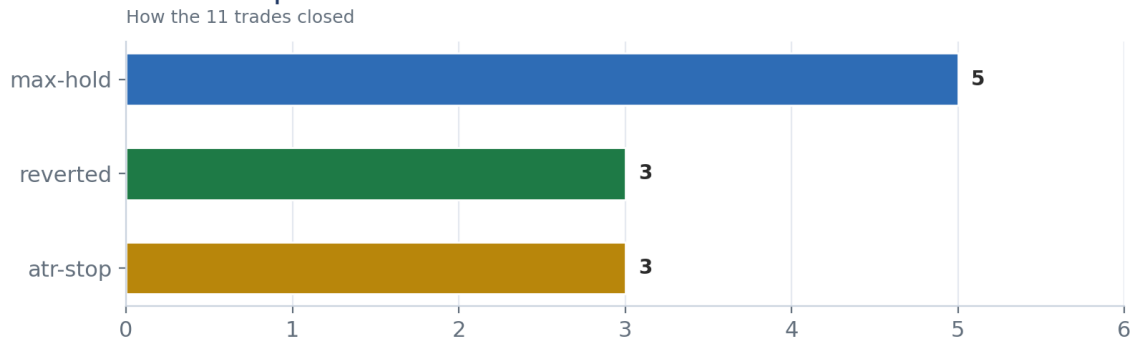


Exhibit 5 — How trades closed: five hit the 30-day cap, three mean-reverted, three stopped out.

7 Research Iteration History

Every configuration tested across the project is reported below, in order, with honest results. Showing what did *not* work — and why — is stronger evidence of a real research process than presenting only a final number. Iterations 1–7 used a 2020–2024 sample; iteration 8 (this report) extends to 2015–2024 for statistical power.

Exhibit 4 | Research Iteration History

Eight configurations tested; iteration 3's higher Sharpe rested on only 3 trades, so it was rejected



#	Configuration	Sharpe	Trades	Verdict
1	Baseline: static hedge, fixed z-bands, no regime filter	~0	n/a	Naive; no edge
2	Rolling hedge + adaptive thresholds + sizing	-0.90	14	Worse; overtrading
3	+ rolling cointegration ($p < 0.10$) + correlation filter	0.58	3	Too few trades
4	Same engine across a semiconductor basket	-0.33	6	Diversification hurt
5	Single cointegration filter, threshold loosened	-0.78	9	Let in weak regimes
6	+ momentum risk-scaler on the shorted leg	-0.75	9	No improvement
7	Cointegration threshold via sensitivity grid	-0.31	5	Method sound, sample short
8	Iteration 7 method, extended 2015–2024 sample	0.44	11	Selected version

Table 5 — Eight iterations. Iteration 3 posted a higher Sharpe (0.58) but on only three trades, so the pre-committed ≥ 5 -trade rule rejected it in favor of iteration 8.

8 Limitations & Future Work

LIMITATIONS (STATED, NOT HIDDEN)

- **Single pair, modest trade count.** Eleven completed trades over ten years is a small sample; results should not be extrapolated to other pairs or eras without re-testing.
- **Return distribution is fat-tailed and skewed** (skew +2.78, excess kurtosis 25.5); Sharpe understates tail risk, and a few trades drive much of the return.
- **Sortino metric needs correction.** The notebook's Sortino (0.17) falls below Sharpe (0.44), which is inconsistent for a positively skewed series — it appears downside deviation was computed over active days only while Sharpe used all days. It is omitted here until recomputed over the identical return series.
- **No slippage or borrow modeling** beyond a flat 7 bps; short-availability and market-impact are assumed away.

Future work

Kalman-filtered hedge ratio for smoother adaptation; a broader screened universe of candidate pairs with the same regime gate; an explicit volatility-regime filter; and walk-forward validation with the cointegration relationship re-qualified on a rolling out-of-sample basis.

BOTTOM LINE

A rigorously constructed, regime-aware pairs trade on AMD/NVDA produces a modest positive risk-adjusted return (Sharpe 0.44) precisely because it refuses to trade a relationship that holds only intermittently. The contribution is the process and its honesty, not the magnitude of the return.

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Full code & notebook: scan the QR or visit the [GitHub repository](https://github.com/edricbinagi/amd-nvda-pairs-trading).

